ZAMBIA UNIVERSITY COLLEGE OF TECHNOLOGY

ICT 1101 Introduction to Programming

Final examination - Academic year 2023, semester 1

Time allowed: 3 hours Total marks: 100 Lecturer: Mr B. Kalaba

Instructions: answer ALL questions. Write clearly. Show all working.

Question 1: Arrays

(a) In most languages, the index of the first array element is:

(b) Accessing index 10 of an array of length 10 causes:

(c) A two-dimensional array is best described as:

(d) What is the index of the last element in an array of length n?

Question 2: Control structures

(a) Which loop always executes its body at least once?

(b) Which statement exits a loop immediately?

(c) How many times does the loop for (i = 0; i < 5; i++) run?

(d) Which structure chooses between many fixed values of one variable?

Question 3: Problem solving

(a) A step-by-step procedure for solving a problem is an:

(b) Which diagram uses a diamond for a decision?

(c) Pseudocode is:

(d) An error that makes a program produce wrong output without crashing is a: